ISYE 6414
Fall 2026

Impact of Environmental Conditions on Global Tourism

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The Research Question

Which average and extreme climate conditions are most strongly associated with annual international tourism growth across countries from 2000 through 2024?

Why It Matters

Understanding these relationships could help tourism planners and policymakers identify climate-related risks and make more informed decisions about tourism development.

Our research question focuses on identifying which average and extreme climate conditions are most strongly associated with annual international tourism growth across countries from 2000 to 2024. We wanted to explore this relationship because climate conditions may influence how attractive a destination is to international tourists. For example, changes in temperature, precipitation, or extreme weather events could affect tourism patterns.

Understanding these relationships could be useful for tourism planners and policymakers when considering climate-related risks and long-term tourism development. Our goal is to determine which climate variable show the strongest associations with tourism growth, rather than assuming that climate conditions directly cause changes in tourism.

Prior Work

Research Gap

How strong is the relationship between specific climate conditions and tourism growth and has it remained the same? Our study examines average and extreme climate indicators across countries from 2000–2024 to identify which show the strongest associations with tourism growth.

Hamilton et al. (2005)	Jaume, (2014)	Yu et al. (2009)
- Built on the Hamburg Tourism Model (HTM) to simulate how the patterns of international tourism would change throughout the 21st century based on several factors, one of them being climate change - Concluded that while a relationship exists, economic standing and population projections had a greater influence on tourism patterns than climate change.	- There are existing "controversies regarding the weaknesses of statistical models in predicting tourist flows under scenarios of climate change" - States that "[the extent] to which tourism has been overlooked in literature on climate change could be explained by the uncertainty and complexity of expected tourism demand reactions."	- Performed an analysis under a similar premise to ours by using hourly "weather data from contrasting climatic regions, Florida and Alaska." - Surveys indicated that 91% of respondents select their destinations based on more attributes than just temperature, "for example, the non-thermal elements of rain, high wind, and low visibility have considerable impact on tourists' satisfaction."

References on Slide 13

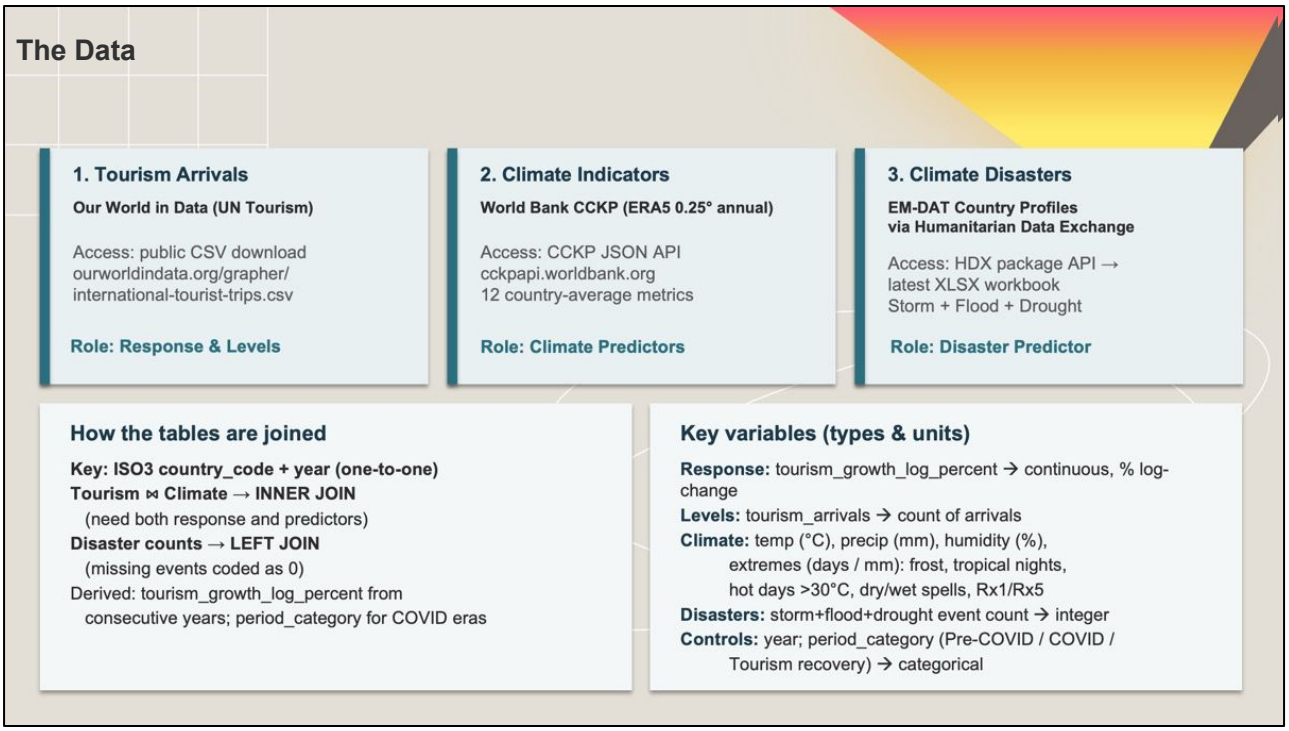
A gap this research question can answer is whether the strength of the relationship between climate patterns and tourism change over time or by region. It is a common assumption that tourism is impacted by weather not only in the physical ability to travel, but also in the attractiveness of a destination. How strong is that relationship and has it remained the same? Specifically, our project examines average and extreme climate conditions across countries from 2000 to 2024 to identify which variables have the strongest associations with annual international tourism growth.

Hamilton et al. (2005) and Jaume, (2014) have already explored the existence of a relationship between our dependent variable and our predictor (climate), but both have also regarded their concern with the lack of attention that has been placed towards understanding the patterns because it is either considered too weak to be statistically significant or because it's too complex to model. This signals a potential gap in the understanding of how the relationship works as well as how it can change.

Yu et al. (2009) supports our team's approach of using individual climate indicators, such as, temperature, precipitation, wind speed, and humidity rather than climate change as a broad concept. It also aligns with our EDA results (scatter plot on slide 10) where we output a Pearson Coefficient of only

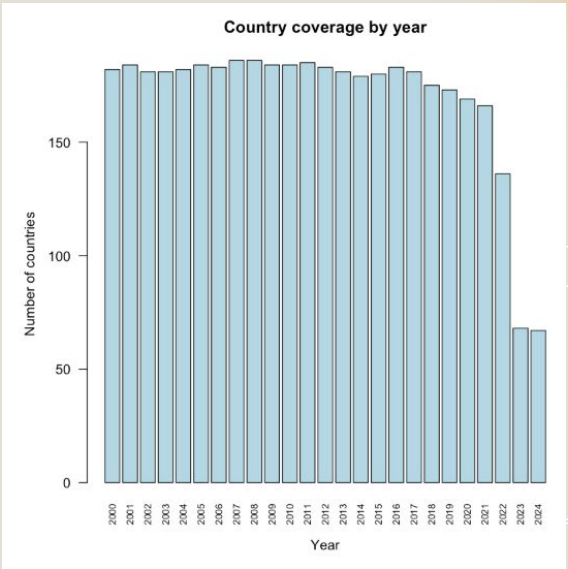
-0.026 for the relationship between average temperature and tourism growth.

These studies provide a foundation for our research, but we also want to explore whether these relationships differ across time periods, particularly given the disruptions to international tourism during COVID-19.



The data ranges from the year 2000-2024 with a total count of 4243 rows with 19 columns (13 predictors). This data set meets the criteria of greater than 3000 rows with 10+ predictors. Lastly, the dataset covers 205 countries.

Exploratory Analysis



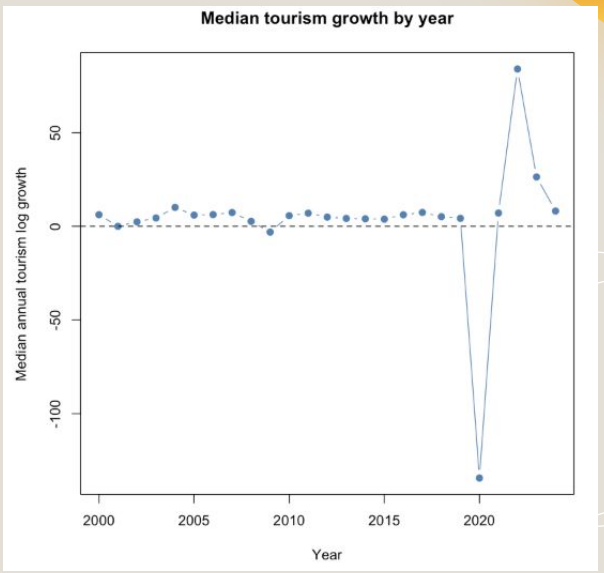
Country coverage drops sharply after 2022, which may affect comparisons of tourism growth across recent years.

This chart shows how the number of countries represented in our dataset varies by year. Country coverage remains relatively consistent for much of the period but declines in recent years. For example. coverage drops from 136 countries in 2022 to 68 in 2023 and 67 in 2024.

Since the climate dataset contains 246 countries each year, the decrease is not due to climate coverage alone. It may also reflect missing or unavailable tourism data for recent years.

This is important because differences in the countries included each year may affect our analysis of tourism growth. If certain countries are missing, the observed trends may not fully represent global tourism patterns, especially in 2023 and 2024.

Exploratory Analysis

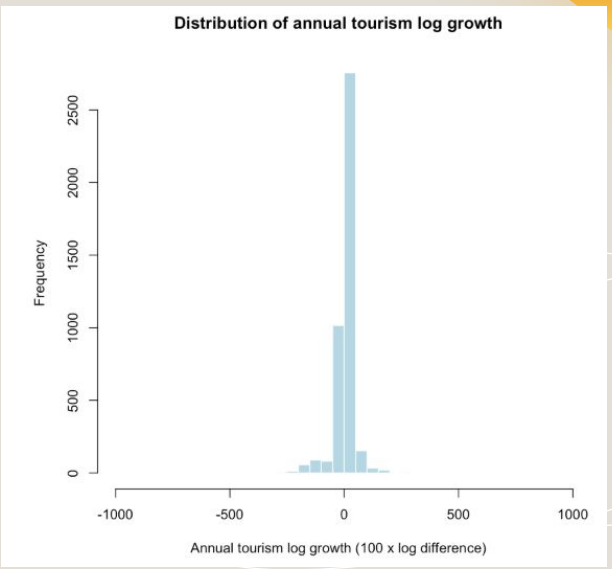


Tourism growth remained relatively stable before 2020, followed by a sharp decline during COVID-19 and a strong recovery in 2022.

This chart shows how the median tourism growth across countries changed from 2000 to 2024. Before 2020, median tourism growth remained relatively stable, with small fluctuations from year to year. However, we see a sharp decline in 2020, followed by a strong rebound in 2022.

These changes align with the COVID-19 pandemic, which disrupted international travel, and the subsequent recovery of the tourism industry. One important consideration is that fewer countries have tourism data available for 2023 and 2024, as we saw in the previous slide. This makes comparisons across recent years less reliable because the countries represented may differ.

Exploratory Analysis

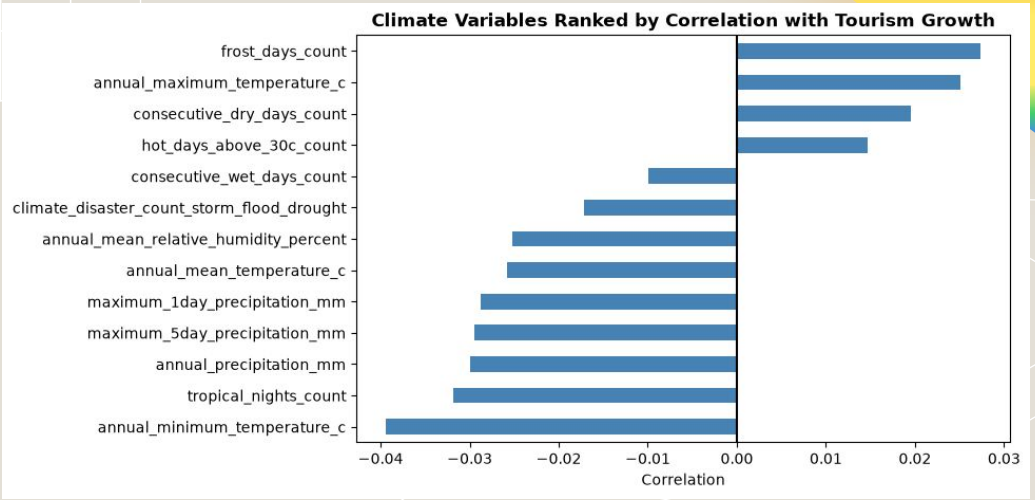


Most tourism growth values cluster near zero, while extreme observations may influence regression results.

This histogram shows the distribution of annual international tourism growth across our dataset. Most observations are concentrated near zero, suggesting that tourism growth was relatively moderate for most country-year observations. However, we also identified some extreme values ranging from approximately −961 to 925. These extreme observations stretch the horizontal axis, making it difficult to see the distribution of the majority of the data.

These outliers are important because they could have a substantial influence on our analysis. Some may reflect major disruptions to tourism, such as COVID-19, while other could be related to unusually large changes in tourism arrivals or potential data-quality issues.

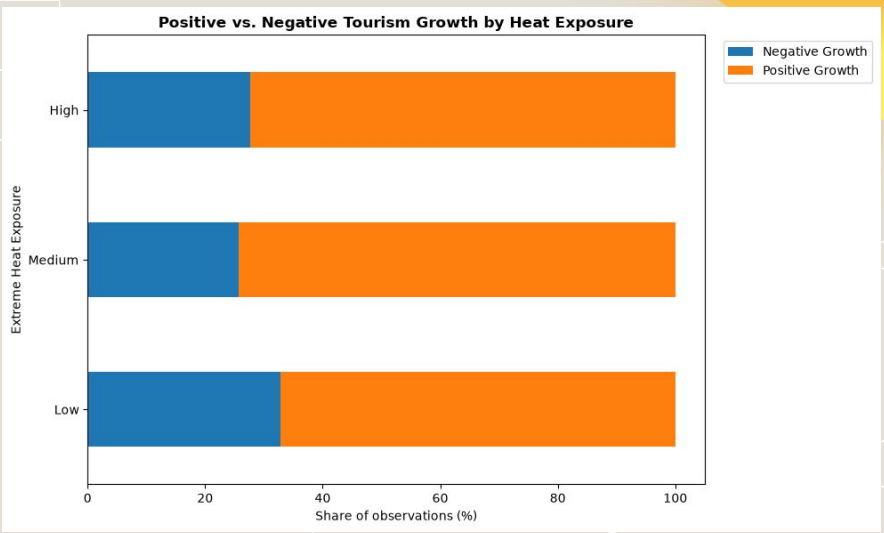
Exploratory Analysis



All climate indicators show weak linear correlations with tourism growth (approximately -0.04 to +0.03).

All climate variables exhibited very weak correlations with tourism growth, with correlation values falling between approximately -0.04 and +0.03. Frost days, annual maximum temperature, and consecutive dry days showed the strongest positive associations, whereas annual minimum temperature, tropical nights, and precipitation measures showed the strongest negative associations. The small magnitudes suggest that climate conditions alone explain little of the variation in tourism growth across countries.

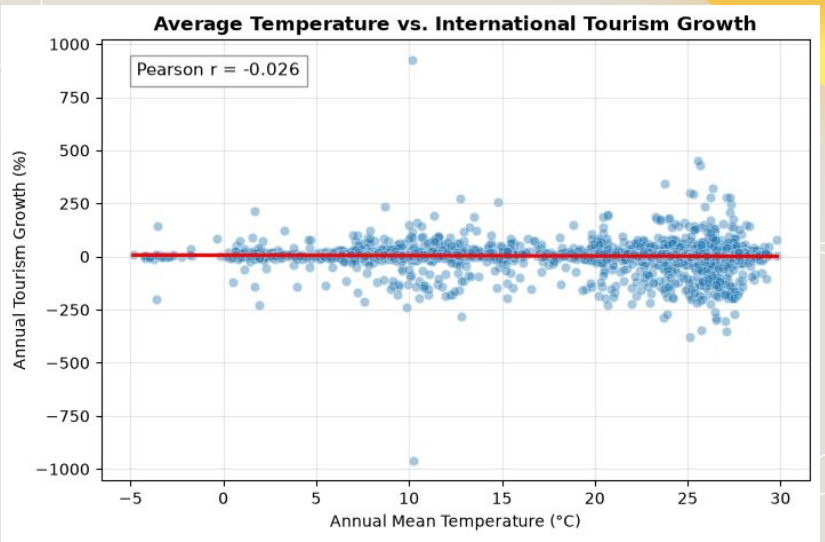
Exploratory Analysis



Positive tourism growth is more common across all heat exposure levels, with the lowest proportion observed in the low-exposure group.

The chart above indicates that positive tourism growth is more common than negative growth across all levels of heat exposure. Although small differences exist among the groups, the overall distribution remains relatively consistent, with approximately 70-75% of observations exhibiting positive growth regardless of heat intensity. These findings suggest that extreme heat exposure alone does not appear to be strongly associated with tourism growth, and that additional economic, geographic, and policy factors are likely influencing tourism performance.

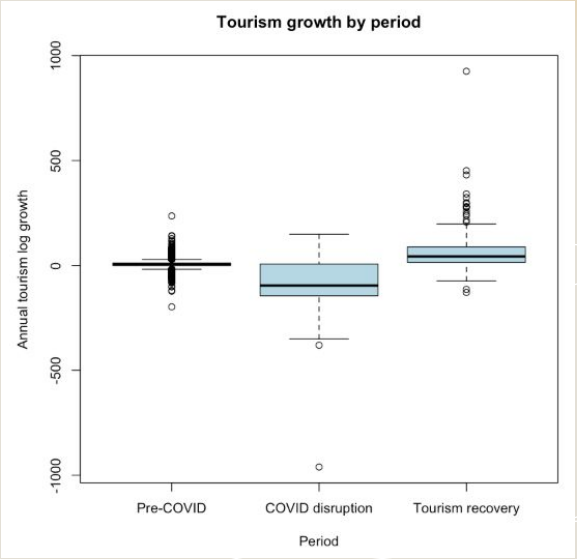
Exploratory Analysis



Average annual temperature has a very weak linear correlation with tourism growth (r = -0.026), although tourism growth appears more variable at higher temperatures.

This scatter plot shows the relationship between average annual temperature and international tourism growth. Annual mean temperature shows little linear relationship with international tourism growth. The Pearson correlation coefficient is r = -0.026, which is a negligible negative correlation and the nearly flat regression line supports this finding. Overall, while tourism growth varies substantially across observations, this variation does not appear to be systematically explained by temperature alone.

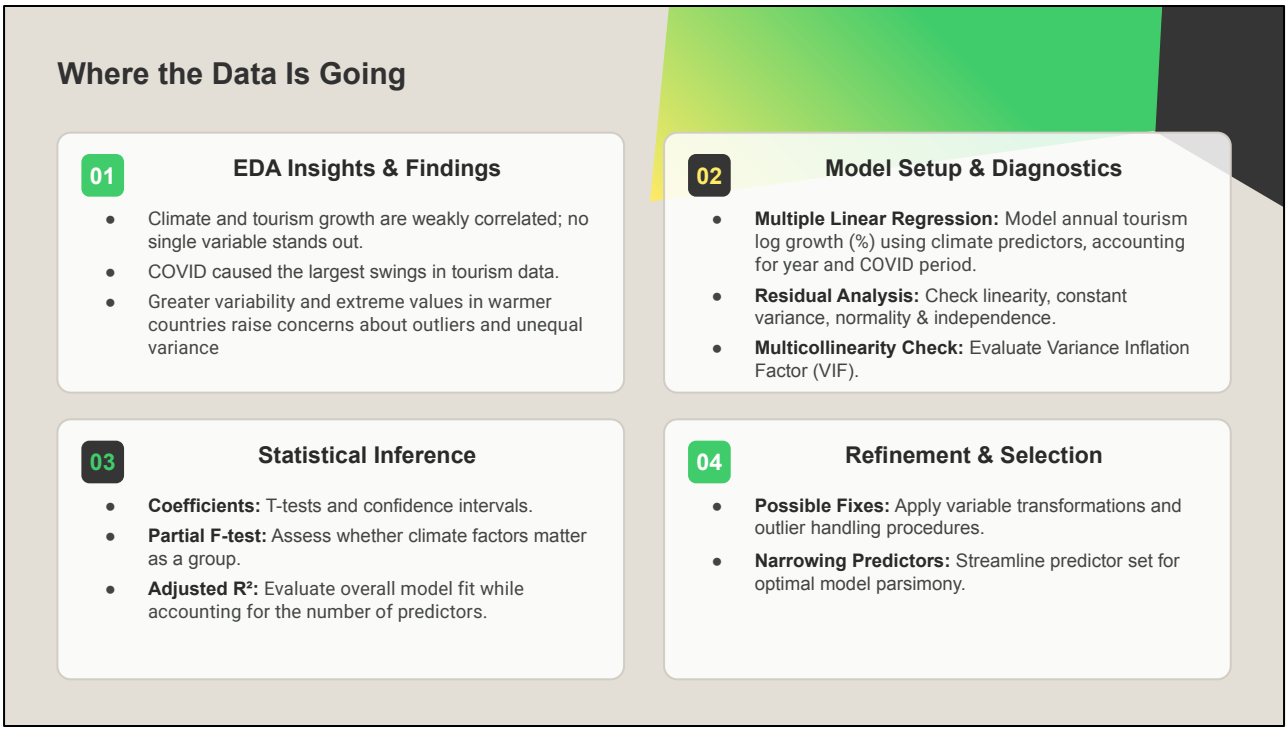
Exploratory Analysis



Tourism growth declined sharply during COVID and rebounded during recovery, with greater variability and extreme values compared to the pre-COVID period.

Median tourism growth was 4.87 before COVID, −95.75 during COVID, and 42.93 during recovery. Tourism growth also varied more during COVID and recovery, with several extreme values.

These differences suggest that the time period is important to consider when analyzing the relationship between climate conditions and tourism growth. The COVID-19 disruption and subsequent recovery may have influenced tourism growth independently of climate conditions, so these changes should not be attributed to climate alone.



From the exploratory data analysis, we saw that climate variables have very weak correlations with tourism growth, ranging from approximately -0.04 to +0.03. That suggests a subtle signal, not necessarily no relationship. Tourism growth shows its largest observed changes during the COVID disruption and recovery periods, which suggests that the time period is an important factor when analyzing the relationship between climate conditions and tourism growth. Moreover, tourism growth is wider spread in warmer countries, which flags outliers and non-constant variance.

We'll start with multiple linear regression, possibly controlling for period and year, so each climate coefficient is estimated net of COVID. Given the weak correlations observed during our EDA, we may expect a low R-squared, but our goal is to determine whether any climate variables show statistically significant associations with tourism growth after accounting for other factors. We'll use residual and Q-Q plots to check linearity, constant variance, normality, and independence. Repeated observations per country are a concern. If assumptions fail, we'll transform, handle outliers, or use robust or clustered standard errors. We'll also check VIFs, since the temperature metrics overlap. To narrow down our predictors, we will use AIC/BIC or LASSO to drop redundant or uninformative variables. For statistical inference, we'll run t-tests, confidence intervals, and a partial F-test for climate as a group. Interactions between climate and period will test whether the relationship has changed over time, which addresses the gap from our prior work.

Overall, the EDA suggests a weak but possibly real relationship, and our next step is to model it carefully and see what the data supports.

References

Hamilton, J., Maddison, D., & Tol, R. (2005). Effects of climate change on international tourism. *Climate Research*, *29*, 245–254. https://doi.org/10.3354/cr029245

Jaume, R.-N. (2014). How to evaluate the effects of climate change on tourism. *Tourism Management*, *42*, 334–340. https://doi.org/10.1016/j.tourman.2013.11.006

Yu, G., Schwartz, Z. & Walsh, J.E. A weather-resolving index for assessing the impact of climate change on tourism related climate resources. *Climatic Change* **95**, 551–573 (2009). https://doi.org/10.1007/s10584-009-9565-7